

Estimating the Temperature of Quantum-Annealer Samples for Neural-Network Variational Monte Carlo

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A quantum annealer can supply the samples needed to train a neural-network wavefunction, but only if its device- and problem-dependent effective temperature is known. For a restricted Boltzmann machine (RBM), the hidden units are conditionally independent given the visible ones, with conditional means fixed by the temperature. We fit these conditional means over all visible–hidden pairs that training already draws, which gives one effective-temperature estimate per training step, without additional device calls or the partition function. Against a known temperature, the estimate has a typical relative error of 5% but fails by more than a factor of three in 2.5% of fits, all at low temperature on trained networks. Used for feedback during training, it reduces the Kullback–Leibler divergence between the annealer samples and the Born distribution by a factor of 4 (Pegasus) to 10 (Zephyr) relative to a heuristic adjustment rule. The samples nevertheless remain further from the Born distribution than those of classical samplers, partly because the feedback rule we used converges to the wrong temperature. As a consequence of this distortion, the energy estimated from annealer samples is biased low, which we confirm by independent re-evaluation at $N = 32$. For the one-dimensional transverse-field Ising model at $N = 64$, two D-Wave annealers and an FPGA sampler reach an energy error of 0.1 per spin within a factor of two of each other in device time, while Metropolis sampling and simulated annealing take more than ten times longer.

I. INTRODUCTION

Variational Monte Carlo (VMC) approximates a ground state by iteratively optimizing a parametrized trial wavefunction, using energies and gradients estimated from configurations sampled from its Born distribution $|\psi_\theta(v)|^2$. Restricted Boltzmann machines (RBMs) are a common choice of trial wavefunction because their amplitudes and logarithmic derivatives are available in closed form [1, 2]. Each optimization step requires a fresh set of samples, usually generated by Markov-chain Monte Carlo.

A quantum annealer can generate these samples instead. The RBM is mapped to an Ising Hamiltonian on the device, and each anneal returns one spin configuration [3]. The annealer output, however, is at best approximately a Boltzmann distribution, at an effective temperature that depends on the device and the problem instance [4, 5]. If this temperature differs from the target, the samples follow the wrong distribution, and the energy and gradient estimates are biased. Every optimization step therefore needs an estimate of the effective temperature, ideally without additional device calls and without the partition function, which is intractable for large systems.

a. Contribution relative to prior work. Kubo and Goto introduced conditional expectation matching (CEM): given a visible configuration, the conditional means of the RBM hidden units are known functions of the temperature, which can therefore be fitted from sampled hidden units [6, 7]. Their method fixes one visible configuration and queries the sampler repeatedly at it. We instead pool all visible–hidden pairs that training already draws, treating each sampled visible configuration as its own condition, and fit a single temperature per step. This requires no additional device calls. Goto and Ohzeki also calibrate annealer parameters online from training samples [8], and Kim *et al.* report faster RBM training with temperature-corrected annealing than with classical sampling [9]. Our contribution is the pooled estimator, a quantitative test of what it does and does not measure, and a benchmark with its conditions stated explicitly.

b. Results.

1. Against a known temperature, the pooled estimate has a typical relative error of 5%, with rare large failures at low temperature (Sec. III). It constrains only the hidden-unit conditionals, not the visible marginal.
2. Feedback with this estimate brings the annealer samples substantially closer to the Born distribution, but not as close as classical samplers. Part of the remaining gap is caused by the feedback rule used in

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our runs, which converges to the wrong temperature (Sec. IV).

3. The energy estimated from the distorted annealer samples is biased low relative to the variational energy of the trained network, which we verify by independent re-evaluation at $N = 32$ (Sec. V).
4. For the one-dimensional transverse-field Ising model at $N = 64$, the annealers reach a fixed energy tolerance in the same device time as an FPGA sampler, within a factor of two, and more than ten times faster than Metropolis sampling and simulated annealing (Secs. V and VI).

II. SETUP

a. Model. We study the transverse-field Ising chain with periodic boundary conditions,

$$H = - \sum_{i=1}^N \sigma_i^z \sigma_{i+1}^z - h \sum_{i=1}^N \sigma_i^x, \quad (1)$$

whose exact ground-state energy E_0 is known [10, 11]. Unless stated otherwise, $h = 0.5$.

b. Wavefunction. An RBM with N visible spins $v_i = \pm 1$, $M = N$ hidden spins $h_j = \pm 1$, biases a, b and couplings W defines the amplitude

$$\psi_\theta(v) \propto e^{-\frac{1}{2} \sum_i a_i v_i} \prod_{j=1}^M [2 \cosh \phi_j(v)]^{1/2}, \quad (2)$$

with $\phi_j(v) = b_j + \sum_i W_{ij} v_i$. The Born distribution $\pi_\theta(v) = |\psi_\theta(v)|^2 / Z_\theta$ is the visible marginal of the Boltzmann distribution $p_\beta(v, h) \propto e^{-\beta \mathcal{E}_\theta(v, h)}$ at $\beta = 1$, with energy

$$\mathcal{E}_\theta(v, h) = \sum_i a_i v_i - \sum_j b_j h_j - \sum_{i,j} W_{ij} v_i h_j. \quad (3)$$

Given v , the hidden units are independent, with conditional means

$$\mathbb{E}_\beta[h_j | v] = \tanh(\beta \phi_j(v)). \quad (4)$$

Parameters are optimized by stochastic reconfiguration (SR) [2, 12] with learning rate 0.08, diagonal regularization 0.05, 200 samples per step and 100 steps.

c. Annealer sampling. We program the Ising Hamiltonian $\mathcal{E}_\theta/\beta_x$, with v and h as qubits, on a D-Wave Advantage_system6 (Pegasus topology) or Advantage2_system1 (Zephyr topology), using a cached minor embedding [3]. If the device output is Boltzmann-distributed at an unknown inverse temperature β_{hw} , the samples follow p_β with

$$\beta_{\text{eff}} = \beta_{\text{hw}}/\beta_x. \quad (5)$$

The target is $\beta_{\text{eff}} = 1$, so the scale factor β_x must be set from an estimate of β_{eff} . Whether the output is Boltzmann-distributed at all is tested in Sec. IV. Each call uses a $20 \mu\text{s}$ anneal, 200 reads and the default chain strength, with automatic coefficient rescaling (`auto_scale`) disabled (Sec. IV).

d. Classical samplers. As baselines we use Metropolis sampling (chains reinitialized every step, 200 warm-up sweeps plus one), persistent block Gibbs sampling (10 sweeps per step), simulated annealing (200 warm-up and 40 cooling sweeps), and an FPGA sampler (Sec. V C). All run at $\beta = 1$ with the same SR settings.

III. POOLED EFFECTIVE-TEMPERATURE ESTIMATE

A. Estimator

At each SR step the annealer returns 200 pairs $(v^{(r)}, h^{(r)})$. By Eq. (4), each sampled $v^{(r)}$ fixes the conditional means of the hidden units as a function of β : close to $\text{sign } \phi_j(v)$ at large β and close to zero at small β . The observed hidden units therefore carry information about the temperature at which they were drawn. We estimate it by least squares over all pairs,

$$\hat{\beta} = \arg \min_{0.01 \leq \beta \leq 50} \sum_{r=1}^{200} \sum_{j=1}^M [h_j^{(r)} - \tanh(\beta \phi_j(v^{(r)}))]^2. \quad (6)$$

This is conditional expectation matching (CEM) [6, 7], pooled over the sampled visible configurations instead of repeated at a fixed one. It uses only samples that training already draws and does not involve the partition function. Its cost is a bounded one-dimensional minimization, a few milliseconds per step.

Figure 1 illustrates the fit. Binning the samples by ϕ and averaging the hidden units gives an empirical conditional-mean curve, which $\tanh(\beta \phi)$ matches for only one β .

[TODO – Check how Fig. 1 was produced. Its script (`cem_matching_demo.py`) samples with the classical Gibbs sampler, which ignores β_x and samples at $\beta = 1$, yet the fit returns $\hat{\beta} \approx 1.94$. Either the figure was made with an older sampler, or the value needs explaining. Regenerate with a known β if in doubt.]

a. Feedback. In our runs the estimate updates the scale factor by an exponential moving average, $\beta_x \leftarrow 0.7 \beta_x + 0.3 \hat{\beta}$, clipped to $[0.05, 20]$. Section IV C shows that this rule does not converge to $\beta_{\text{eff}} = 1$.

B. What the estimate does not constrain

The loss in Eq. (6) depends only on the hidden-unit conditionals. Write $q(v)$ for the sampler's visible marginal

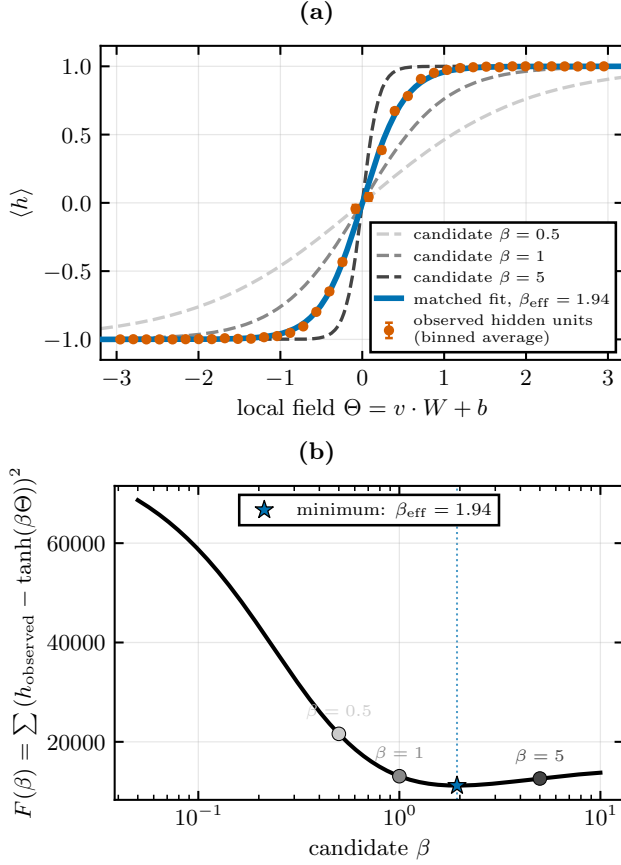


FIG. 1. Pooled CEM fit for an $N = 16$, $h = 1$ network. (a) Binned hidden-unit means against ϕ (points) and $\tanh(\beta\phi)$ for three trial values of β . (b) Least-squares loss of Eq. (6) as a function of β , minimized at $\hat{\beta} \approx 1.94$.

and $m_j(v)$ for its hidden conditional means. The expected loss per sample is

$$\sum_j \mathbb{E}_{q(v)} \left[\text{Var}_q(h_j | v) + (m_j(v) - \tanh(\beta \phi_j(v)))^2 \right]. \quad (7)$$

If $m_j(v) = \tanh(\beta_0 \phi_j(v))$ on the support of q , then β_0 minimizes Eq. (7) for any $q(v)$. A correct $\hat{\beta}$ therefore does not imply a correct visible marginal. For example, take one visible and one hidden spin with $a = 0$, $b = 0.2$, $W = 0.4$ and target $\beta = 1$. A sampler that always returns $v = -1$ but draws h from the correct conditional gives $\hat{\beta} = 1$, while its visible marginal is at total-variation distance $\cosh(0.6)/[\cosh(0.2) + \cosh(0.6)] \approx 0.54$ from π_θ . Section IV therefore measures the visible marginal directly.

TABLE I. Accuracy of $\hat{\beta}$ against β_{true} (Fig. 2), 108 fits per training stage. Fail: fits off by more than a factor of three. Bias: median of $\hat{\beta} - \beta_{\text{true}}$. RMSE with all fits and without failures (excl.). Rel.: median relative error. Slope: linear fit of $\hat{\beta}$ on β_{true} , without failures.

stage	fail	bias	RMSE	RMSE excl.	rel.	slope
early	0/108	+0.005	0.117	0.117	6.3%	1.02
mid	3/108	-0.003	6.57	0.106	5.6%	0.97
late	5/108	-0.001	9.24	0.101	5.3%	0.95

C. Accuracy against a known temperature

To test the estimator against a known value, we take one $N = 8$, $h = 1$ network at three training stages (early, mid, late) and multiply all its parameters by $\beta_{\text{true}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.3, 1.6, 2.0, 2.5\}$. Exact block Gibbs sampling of the scaled network at $\beta = 1$ samples the original network at β_{true} . For each value we draw 200 samples 12 times (324 fits in total) and evaluate Eq. (6) with the unscaled parameters, so that an exact estimator returns β_{true} .

Excluding failures, the estimate is unbiased, with median relative error 5–6% and slope 0.95–1.02 (Table I). In 8 of 324 fits (2.5%) it is off by more than a factor of three, and five of these reach the upper bound $\beta = 50$. All failures occur at $\beta_{\text{true}} \geq 2$ on the mid and late networks, i.e. at low temperature on trained parameters; we have not identified the cause. The moving average in the feedback rule damps an isolated failure but does not remove it.

The test uses one small network and an exact sampler. It measures the statistical accuracy of the estimator when the samples are Boltzmann-distributed, not its behavior on distorted samples of the kind discussed in Sec. III B.

IV. SAMPLE QUALITY WITH FEEDBACK

For $N \leq 16$ the Born distribution π_θ can be computed by exact enumeration of all 2^N configurations. We compare the empirical distribution $\hat{q}$ of a sampler with π_θ using the Kullback–Leibler divergence $D_{\text{KL}}(\hat{q} \parallel \pi_\theta)$ and the total-variation distance $D_{\text{TV}}(\hat{q}, \pi_\theta)$. With 200 samples both are nonzero even for an exact sampler; we give that finite-sample value as reference where available.

A. Automatic coefficient rescaling

By default, D-Wave rescales the programmed coefficients to the full hardware range, which cancels any choice of β_x . Figure 3 shows this for an $N = 8$ network embedded without chains on Pegasus. With rescaling enabled, neither D_{TV} nor $\hat{\beta}$ depends on β_x . With rescaling disabled, D_{TV} decreases from ≈ 0.63 to a minimum of ≈ 0.30

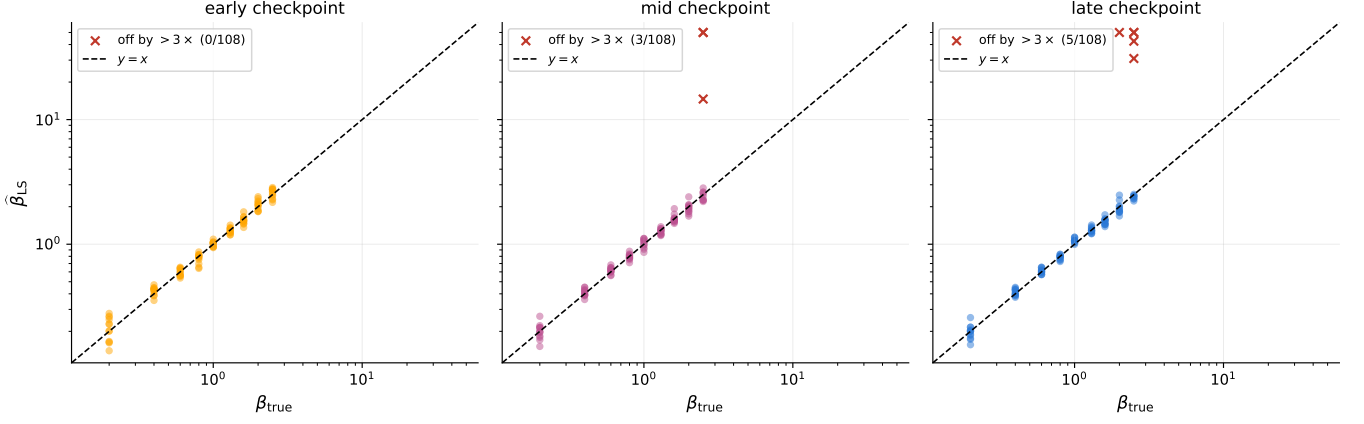


FIG. 2. Pooled CEM estimate $\hat{\beta}$ against β_{true} for an $N = 8$, $h = 1$ network at three training stages (200 exact Gibbs samples per fit, 12 fits per value). Crosses mark fits off by more than a factor of three, all at $\beta_{\text{true}} \geq 2$ on the trained network.

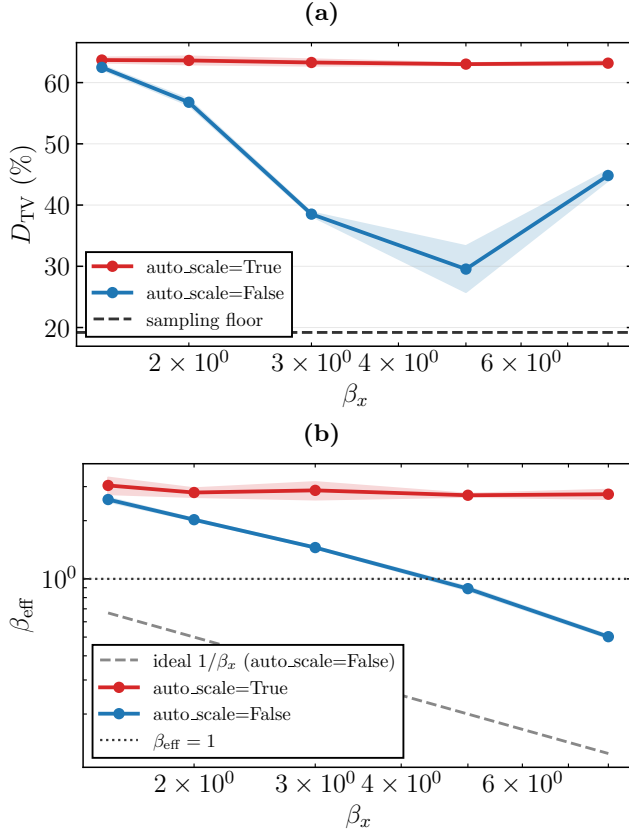


FIG. 3. Automatic coefficient rescaling on Pegasus, $N = 8$. (a) D_{TV} between annealer samples and π_θ as a function of β_x . (b) Pooled CEM estimate $\hat{\beta}$. With rescaling enabled, β_x has almost no effect. The reference line is the finite-sample value of an exact sampler.

near $\beta_x = 5$, compared with a finite-sample reference of ≈ 0.19 . All annealer runs below use rescaling disabled.

B. Effect of CEM feedback

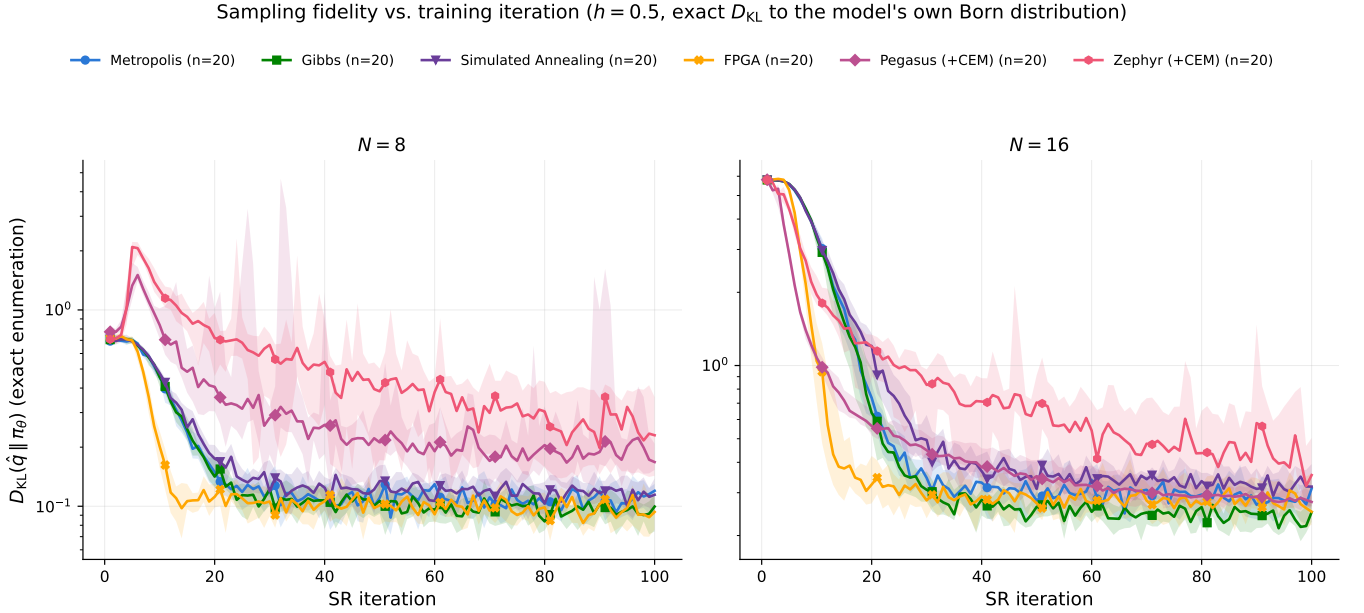
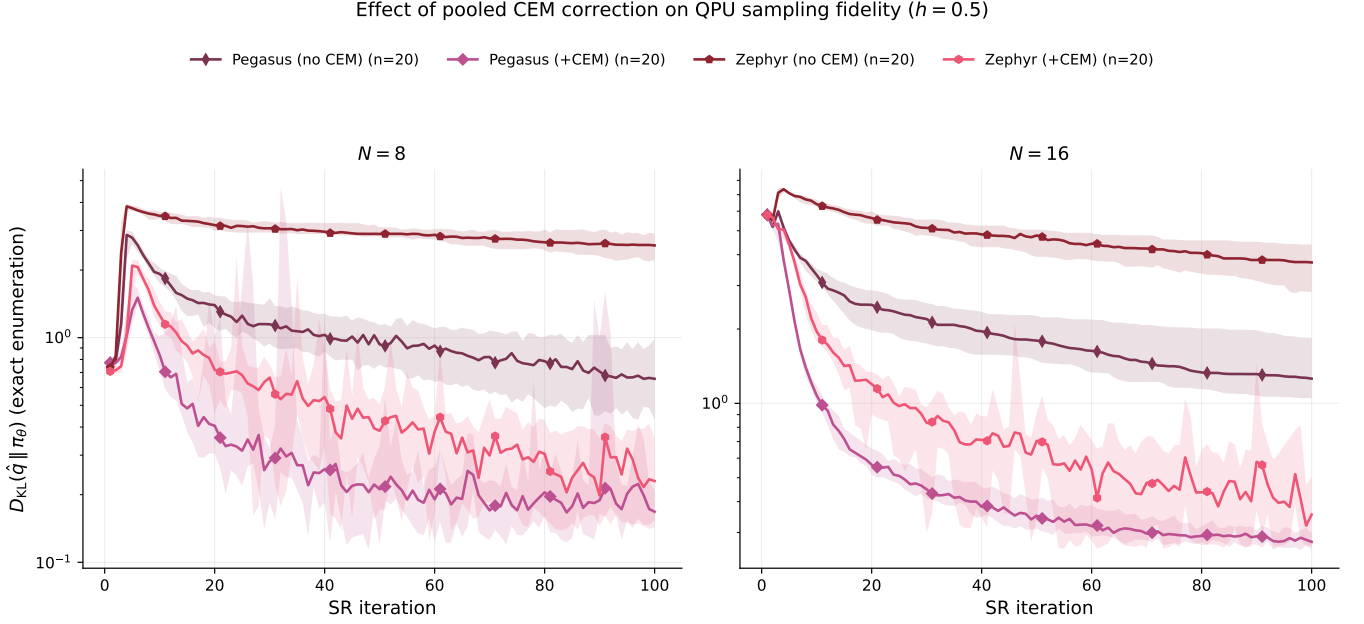
Each training run at $N \leq 16$ records $D_{\text{KL}}(\hat{q} \parallel \pi_\theta)$ at every step, with π_θ the Born distribution of the current parameters. Figure 4 compares annealer runs with CEM feedback against runs with a heuristic rule that multiplies β_x by 1.05 or 0.95, chosen at random, whenever the energy increases. With CEM feedback the final D_{KL} is lower by a factor of about 4 on Pegasus ($0.66 \rightarrow 0.17$ at $N = 8$, $1.26 \rightarrow 0.27$ at $N = 16$) and about 10 on Zephyr ($2.57 \rightarrow 0.23$, $3.72 \rightarrow 0.35$). The difference persists over the whole run.

Figure 5 compares the annealer runs with the classical samplers. Metropolis, Gibbs, simulated annealing and the FPGA converge to a common D_{KL} within 20–30 steps. Both annealers remain above this level, with larger variation between runs. An independent live test agrees: ten Pegasus calls (200 reads each, $\beta_x = 2$) on a trained $N = 8$ network give $D_{\text{TV}} = 0.64$ (interquartile range 0.62–0.64).

C. Fixed point of the feedback rule

Because Eq. (6) is evaluated with the unscaled parameters, $\hat{\beta}$ estimates $\beta_{\text{eff}} = \beta_{\text{hw}}/\beta_x$ (Eq. (5)). The update that yields $\beta_{\text{eff}} = 1$ is $\beta_x \leftarrow \beta_x \hat{\beta}$. The rule we used, $\beta_x \leftarrow 0.7 \beta_x + 0.3 \hat{\beta}$, has its fixed point at $\beta_x = \hat{\beta} = \beta_{\text{hw}}/\beta_x$, i.e. $\beta_x = \sqrt{\beta_{\text{hw}}}$ and $\beta_{\text{eff}} = \sqrt{\beta_{\text{hw}}}$.

We confirmed this with a simulated annealer: exact Gibbs sampling at a prescribed β_{hw} of the network with parameters divided by β_x , inside the unmodified training code ($N = 8$, $h = 0.5$, three runs per value). For $\beta_{\text{hw}} = 3$, β_x converges to 1.75, 1.77 and 1.75 ($\sqrt{3} = 1.73$). For $\beta_{\text{hw}} = 5$ it converges to 2.27 and 2.22 ($\sqrt{5} = 2.24$), and to 4.09 in the third run. In the archived annealer runs, the final β_x is 1.4–3.2 (median over 20 runs, per size and device). At the fixed point $\beta_{\text{eff}} = \beta_x$, so these samples



were drawn at inverse temperatures 1.4–3.2 instead of 1. This is a plausible contributor to the gap in Fig. 5.

Replacing the average by the multiplicative update is not sufficient: in the same simulation it reached $\beta_{\text{eff}} \approx 1$ in 2 of 9 runs, overshoot in the others, and gave larger D_{KL} than the original rule. A stable update needs, for example, damping and rejection of failed fits. The improvement in Fig. 4 is therefore that of a feedback loop that does not

reach the target temperature.

[TODO – Design a stable feedback rule, test it with the simulated annealer, and rerun the annealer experiments of Figs. 4, 5 and 6 with it. Log the raw per-step $\hat{\beta}$; currently only the smoothed β_x is stored.]

TABLE II. Cumulative sampling time (s) at which the median error in Fig. 6 first falls below $\epsilon = 0.1$.

sampler	$N = 8$	$N = 16$	$N = 32$	$N = 64$
Metropolis	0.64	1.41	3.39	11.5
Gibbs	0.65	0.65	0.77	1.04
Simulated annealing	0.94	2.14	5.59	17.1
FPGA	0.007	0.024	0.10	0.59
Pegasus (+CEM)	0.41	0.15	0.50	0.47
Zephyr (+CEM)	0.68	0.53	0.75	0.84

V. BENCHMARK: ENERGY ERROR AGAINST SAMPLING TIME

a. Timing. We count sampling time only. For the annealers this is the reported `qpu_access_time` (programming, annealing and readout) plus the CEM fit, which takes a few milliseconds. For the classical samplers it is the duration of the sampling call on a GPU, excluding one-time JIT compilation. For the FPGA it is the on-device cycle count divided by the 120 MHz clock frequency. The SR update is identical for all samplers and is excluded, as are network latency, queueing and embedding (quantified in Sec. VB).

b. Protocol. For each sampler and $N \in \{8, 16, 32, 64\}$ we run 20 independent trainings with the settings of Sec. II. At step t we estimate the energy from that step’s samples, average it over the last 10 steps to obtain $\bar{E}_t$, and record the error per spin $|\bar{E}_t - E_0|/N$. Figure 6 shows this error against cumulative sampling time.

c. Time to tolerance. Up to $N = 32$ the FPGA reaches $\epsilon = 0.1$ first, 5 to 60 times faster than Pegasus (Table II). The annealer times are nearly independent of N , since the device time per call is set by the anneal time and number of reads; the FPGA time increases with N . At $N = 64$ Pegasus reaches $\epsilon = 0.1$ after 0.47 s, the FPGA after 0.59 s and Zephyr after 0.84 s, with overlapping interquartile ranges. Gibbs follows at 1.0 s, while Metropolis and simulated annealing take more than ten times longer. With 20 runs per sampler, the order of the three fastest samplers at $N = 64$ is not statistically resolved.

d. Final error. Up to $N = 32$ all classical samplers end at a lower error than the annealers (≈ 0.001 against 0.003–0.05 per spin), consistent with the distortion of the annealer samples (Sec. IV). At $N = 64$ the order reverses: the annealers end at 0.008–0.010, the classical samplers at 0.04–0.08. Given the bias shown in Sec. V A, the annealer values at $N = 64$ require independent confirmation.

A. Bias of the sampled energy

For samples drawn from π_θ , the mean local energy is an unbiased estimator of the variational energy $E_\theta = \langle \psi_\theta | H | \psi_\theta \rangle$. For samples from a distorted distribution

$q \neq \pi_\theta$ it is biased, and it can even fall below E_0 , which E_θ cannot by the variational principle.

To measure this bias, we trained Pegasus at $N = 32$ (5 runs, same settings), stored the parameters every 20 steps, and re-evaluated the energy E_{indep} of each stored network with independent, freshly seeded Gibbs chains (95% bootstrap confidence intervals of half-width ≤ 0.0033 per spin). Figure 7 shows the difference to the energy E_{annealer} estimated from the annealer samples at the same step.

The annealer estimate is below the independent confidence interval at all 25 checkpoints. The median bias decreases from -0.083 per spin at step 20 to -0.029 at step 40 and -0.010 at step 99. In two runs the annealer estimate lies below E_0 . At step 20 the bias is comparable to the tolerance $\epsilon = 0.1$, so four of five runs appear to have reached ϵ while the independent estimate shows they have not. From step 40 on the bias is smaller than ϵ but of the same order as the final errors in Fig. 6. Crossing times at $\epsilon = 0.1$ after the first few tens of steps are therefore approximately correct, whereas early crossings and final errors of order 0.01 are not reliable without independent re-evaluation.

[TODO – Repeat the independent re-evaluation at $N = 64$ for both annealers, where the reported final errors are smallest. This requires new annealer runs with stored checkpoints; the benchmark runs did not store them.]

B. Wall-clock time and hardware resources

For ten live Pegasus calls at $N = 8$ the wall-clock time per call was 3.9–10.8 s (median ≈ 5 s), against 31 ms of device time, a ratio of 150–300. The device time splits into programming (15.9 ms) and sampling (15.1 ms for 200 reads with 20 μs anneal and 35 μs readout each). We did not decompose the wall-clock overhead further.

The dense RBM requires minor embedding, in which each logical spin is represented by a chain of physical qubits. Table III lists the embedding cost. Zephyr needs fewer qubits than Pegasus up to $N = 32$, but more qubits and longer chains at $N = 64$, consistent with its larger time to tolerance at that size. In the live $N = 8$ calls the chain-break fraction per read was mostly zero and at most 0.06.

C. FPGA sampler

The FPGA runs a simulated-annealing sampler (VeloxQ) with 1024 parallel replicas in single precision; 200 of the replicas are drawn uniformly at random as the samples of each step. We treat it as a black box and report its sampling time and, for $N \leq 16$, its D_{KL} (Fig. 5), which matches that of the classical samplers.

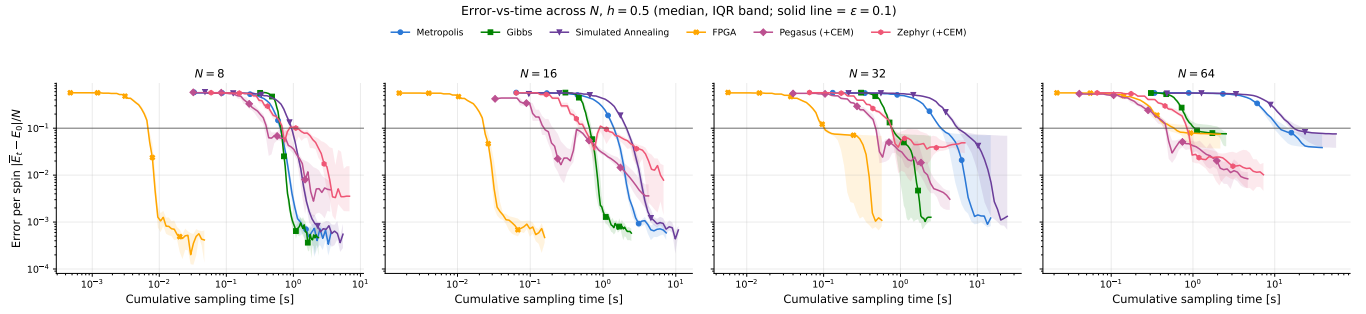


FIG. 6. Energy error per spin (10-step moving average) against cumulative sampling time, $h = 0.5$. Median and interquartile range over 20 runs per sampler and size. The horizontal line marks $\epsilon = 0.1$. Energies are estimated from each sampler’s own samples; for the annealers this estimate is biased low (Fig. 7).

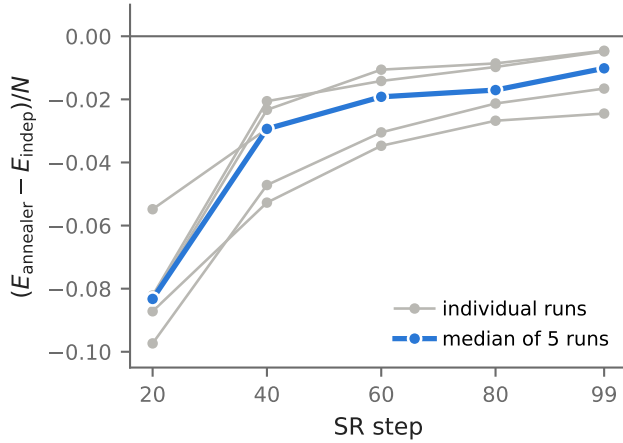


FIG. 7. Bias of the energy estimated from annealer samples, $(E_{\text{annealer}} - E_{\text{indep}})/N$, where E_{indep} is the energy of the same network from independent Gibbs sampling. $N = 32$, Pegasus (+CEM), 5 runs.

TABLE III. Physical qubits and mean chain length of the minor embedding of the dense $N \times N$ RBM.

N	Pegasus		Zephyr	
	qubits	chain	qubits	chain
8	22	1.38	16	1.00
16	64	2.00	48	1.50
32	212	3.31	192	3.00
64	768	6.00	956	7.47

[TODO – Document the FPGA sampler: update rule, stationary distribution and temperature control, board, toolchain and bitstream version, and host-transfer cost. Requires access to the vendor package and hardware.]

VI. LIMITATIONS

- **Feedback rule.** All annealer runs used a rule with fixed point $\beta_{\text{eff}} = \sqrt{\beta_{\text{hw}}}$ instead of 1 (Sec. IV C). Results with a correct rule may differ.
- **Energy bias.** Benchmark errors are estimated from each sampler’s own samples, which for the annealers is biased low. We quantified the bias only at $N = 32$ on Pegasus (Sec. V A).
- **Single model and field.** All benchmark results are for the one-dimensional transverse-field Ising model at $h = 0.5$. Since π_θ changes qualitatively with h , other fields may give a different ordering. The only annealer data at another field are five Pegasus runs at $N = 16$, $h = 1.0$, which reached final errors of 0.027–0.031 per spin.
- **Untuned classical samplers.** Each classical sampler uses a fixed number of sweeps per step. Their times are therefore not optimal; simulated annealing in particular improves with longer schedules.
- **Sample size.** Twenty runs per sampler and size resolve the factor of ten to Metropolis and simulated annealing, but not the order of the three fastest samplers at $N = 64$.
- **Device time.** Wall-clock time per annealer call exceeds device time by a factor of 150–300 (Sec. V B). In wall-clock time the annealers are not competitive in this setup.
- **System size.** Four sizes up to $N = 64$ do not determine the scaling at larger N .

VII. CONCLUSION

The effective temperature of a quantum annealer used as an RBM sampler can be estimated at every training step, without additional device calls, by a least-squares fit of the hidden-unit conditional means over the samples that

training already draws. Against a known temperature the estimate has a typical relative error of 5%, with rare failures at low temperature, and by construction it does not constrain the visible marginal. CEM feedback reduces $D_{\text{KL}}(\hat{q}||\pi_\theta)$ of the annealer samples by a factor of 4 to 10 relative to a heuristic rule, but the samples remain further from π_θ than those of classical samplers, partly because the feedback rule we used converges to $\beta_{\text{eff}} = \sqrt{\beta_{\text{hw}}}$. Energies estimated from these samples are biased low and require independent re-evaluation. Within these limits, for the one-dimensional transverse-field Ising model at $N = 64$, the annealers reach $\epsilon = 0.1$ in the same device time as an FPGA sampler, within a factor of two, and more than ten times faster than Metropolis sampling and simulated annealing.

The next steps are a stable feedback rule, annealer runs with it, independent re-evaluation at $N = 64$, and a benchmark over several fields and tolerances.

DATA AVAILABILITY

Code and data are available in the project repositories [13, 14]. *[TODO: add final commit hashes and a DOI when the manuscript is frozen.]*

USE OF AI ASSISTANCE

An AI assistant (Claude, Anthropic) helped restructure and edit this manuscript. It wrote and ran several re-analyses of archived data, the known-temperature test (Sec. III C) and the simulated-annealer test of the feedback rule (Sec. IV C), and found the error in the feedback rule during code review. The authors checked all results and take responsibility for the content. *[TODO: confirm and state who ran the live annealer calls of Secs. IV, VA and VB.]*

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